ChE 210 Homework Set #10 Due on Friday, 7 November 2019

(1) The standard heat of the reaction

$$CaC2(s) + 5\ H2O(l) \rightarrow.\ CaO(s) + 2\ CO2(g) + 5\ H2(g)$$

is $\Delta\hat{H}_r^\circ = +69.36$ kJ/mol.

(a) Is the reaction exothermic or endothermic at 25°C? Would you have to heat or cool the reactor to keep the temperature constant? What would the temperature do if the reactor ran adiabatically? What can you infer about the energy required to break the molecular bonds of the reactants and that released when the product bonds are formed?

(b) Calculate $\Delta\hat{U}_r^\circ$ for this reaction. Briefly explain the physical significance of your calculated value.

(c) Suppose you charge 150.0 g of CaC2 and liquid water into a rigid container at 25°C, heat the container until the calcium carbide reacts completely, and cool the products back down to 25°C, condensing essentially all the unconsumed water. Write and simplify the energy balance equation for this closed constant-volume system and use it to determine the net amount of heat (kJ) that must be transferred to (or from) the reactor. Indicate the direction of heat transfer.

(2) Murphy 6.32 N.B. Consider the change in energy in the inner combustion chamber as being completely due to the chemical reaction and ignore the sensible energy change of the $C_8H_{10}O_3$ and its products.

(3) Murphy 6.34

(4) Use tabulated heats of formation to determine the standard heats of the following reactions in kJ/mol, letting the stoichiometric coefficient of the first reactant in each reaction equal one.

(a) Nitrogen (N_2) + oxygen (O_2) react to form nitric oxide (NO) al in the gas state.

(b) Gaseous *n*-pentane + oxygen react to form carbon monoxide + liquid water.

(c) Liquid *n*-hexane + oxygen react to form carbon dioxide + water vapor. After doing the calculation, write the stoichiometric equations for the formation of the reactant and product species, then use Hess's law to derive the formula you used to calculate $\Delta\hat{H}_r^\circ$.

(d) Liquid sodium sulfate + carbon monoxide react to form liquid sodium sulfide + carbon dioxide. [The sodium compounds are not listed in the textbook tables. You will need to find data elsewhere.]

(5) Murphy 6.37